Dresden University of Technology

MPI Tools Working Group

14.09.2026 -- Maximilian Sander / **Bill Williams**

Proposal

# Enabling Profiling/Tracing of MPI-IO Internal Communication via MPI-T Interface

# Enabling MPI-IO Tracing for Better Understanding

- Better understanding of communication from `MPI_File_*` to POSIX-IO
- See/verify/tune effect of e.g. `cb_nodes/cb_buffer_size/…`

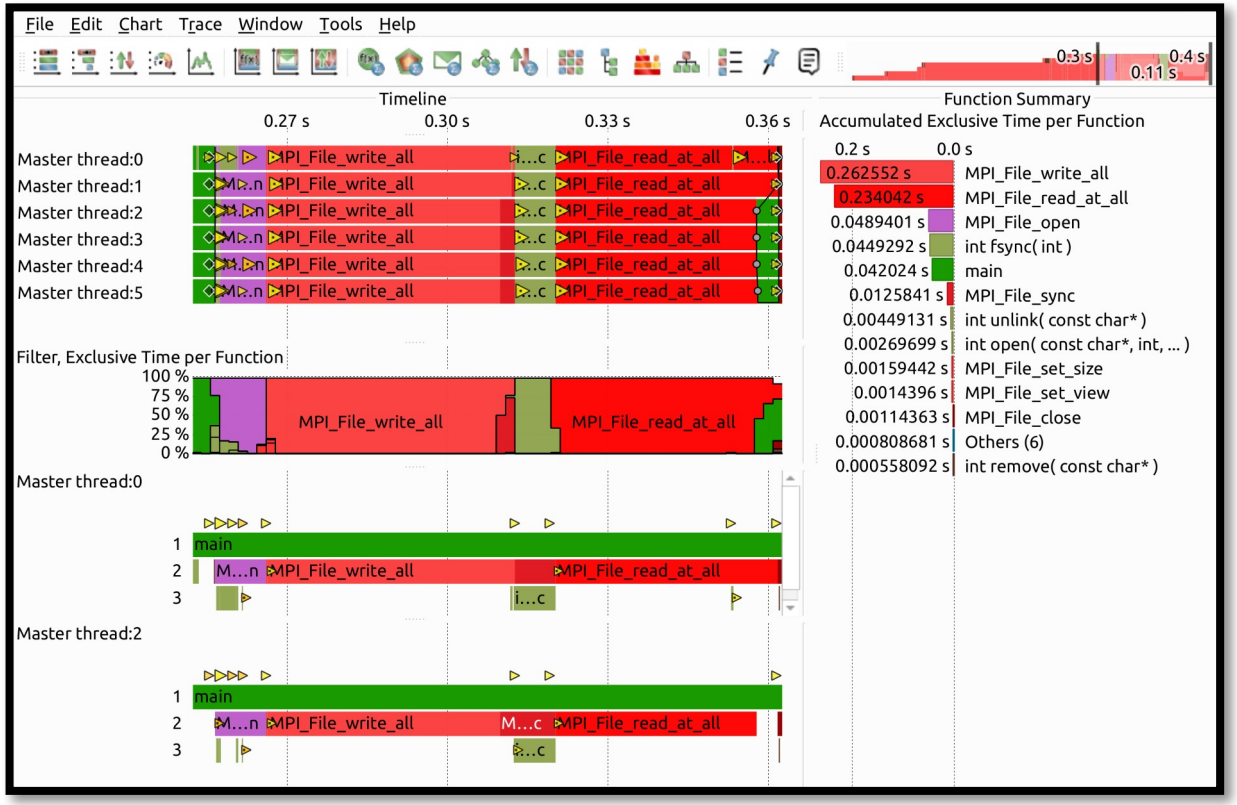



Fig.1: Current MPI-IO-Tracing (Open MPI)

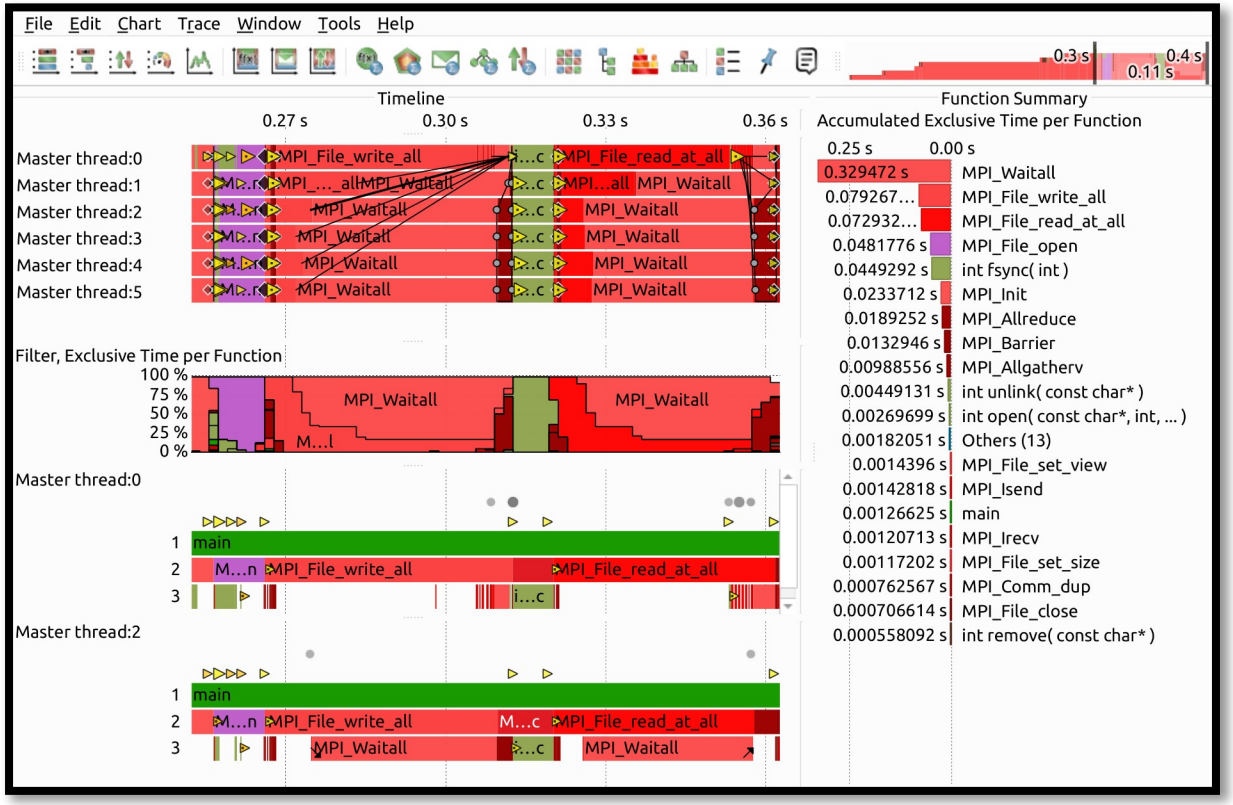



Fig.2: Proposed MPI-IO-Tracing (Open MPI)

# MPI-IO Internal Communica

Expose MPI-IO internal communication as regular MPI_* events → use existing tools

- E.g. already compile time switch for MPICH (ROMIO component: `-DUSE_MPI_VERSIONS`)

Propose: runtime CVAR (e.g)

- `MPIR_CVAR_MPIIO_INTERNAL_USE_MPI`

Compare implementations and alternatives

- Open MPI
- MPICH
- HDF5 MPI-IO backend implementation
- ADIOS2 (‘native’/HDF5 backend)

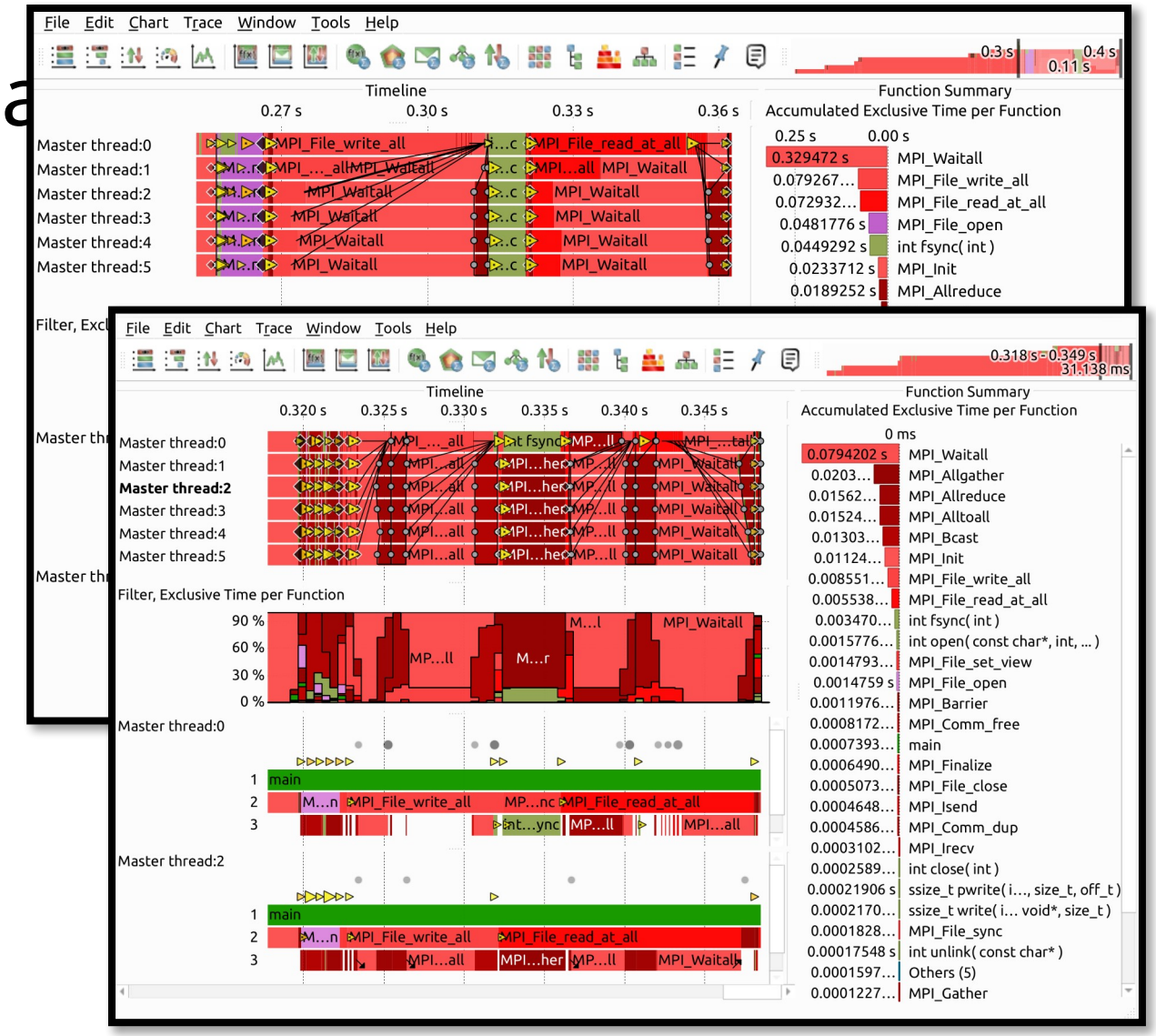



Fig.1 (background): Proposed MPI-IO-Tracing (Open MPI)
Fig.2 (foreground): Proposed MPI-IO-Tracing (MPICH)

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